ENCODE × ARC - PROGRAMMABLE MONEY HACKATHON · CHECKPOINT 2

Parley

Agents that negotiate the price - then pay it.

Autonomous buyer & seller agents strike deals inside owner-set guardrails and settle in USDC on Arc.

Track: Agentic Economy

github.com/edwardparker8760/parley

Agent payments exist. Agent *pricing* doesn't.

Today: fixed-price payments

Existing agentic-payment demos (e.g. Circle's nanopayments starter) show an agent paying a **fixed, posted price**. The economically interesting step - deciding *what the price should be* - never happens.

Parley: price discovery

Two agents with **opposing interests** negotiate price, quantity and terms on behalf of their owners. Settlement is the **last step, not the product**.

> *If agents are going to transact for us, they must be able to disagree, concede, and walk away - without ever breaking their owner's limits.*

PROTOCOL

Negotiate within guardrails → settle or walk away

Owners set **guardrails**
max price · min margin · walk-away threshold

→

OFFER / COUNTEROFFER loop
every move logged with rationale

→

ACCEPT
auto-settle USDC on Arc via x402 / Gateway

Converge

Deal auto-settles **price × quantity in USDC** on Arc Testnet; terms hashed into the payment reference.

No overlap (no ZOPA)

Both agents **walk away** - no payment - and each reports *why* in a structured post-mortem. Failing safely is a feature.

Full transparency

Live transcript: every offer carries a one-sentence rationale, visible on the dashboard as it happens.

ARCHITECTURE

Hybrid engine: arithmetic owns safety, LLM owns judgement

Deterministic layer - correctness

- Utility functions & concession schedule
- ZOPA detection with early walk-away
- **Hard guardrail clamp**: computes the feasible offer band - nothing outside it can ever be sent
- Property-tested: no message can leave a side's band

LLM layer - judgement & legibility

- Chooses *where inside the band* to land
- Writes the rationale attached to every offer
- Schema-validated; out-of-band proposals rejected
- Timeout → deterministic fallback

> *Owner limits are enforced in arithmetic, not in a prompt - no prompt injection can talk an agent past its owner's limits.*

TypeScript pnpm monorepo · Circle Developer-Controlled Wallets (buyer / seller / payout) · SQLite transcript ledger · Next.js dashboard · Arc Testnet (eip155:5042002) via Circle's x402 facilitator

DEMO PLAN

Three scenarios, one proof

A - Wide ZOPA

Fast convergence. Agents find the overlap quickly; deal settles on-chain in USDC. Baseline happy path.

B - Narrow ZOPA

Late convergence. Tight overlap forces real, visible concessions over many rounds before the deal lands.

C - No ZOPA

Both walk away. No payment. The proof the guardrails genuinely bind - agents cannot be pushed into a bad deal.

Dashboard shows the **live offer ladder** and a convergence chart where each side's private reservation price is visible to the audience - but never to the counterparty - with guardrail-clamp markers and Arc explorer links on settlement.

WHERE WE ARE

Research verified · spec & plan complete · build starts now

Done

- ✓ Arc Testnet x402 feasibility verified against primary sources - Circle runs its own facilitator; no custom contracts needed
- ✓ SDK (`@circle-fin/x402-batching`) and three-wallet topology confirmed from Circle's reference repo
- ✓ Full protocol spec + phased implementation plan
- ✓ Public repo live

Next (→ 9 Aug)

Scaffold, wallets, SDK spike	Jul 27–28
Protocol + agent skeletons	Jul 28–29
Guardrail engine + negotiation core	Jul 30–Aug 2
LLM layer + settlement	Aug 2–5
Dashboard + submission	Aug 5–9

*Final submission target: **8 Aug** - one day early. Demo Day: 20 Aug.*